

Logarithm Formulas & Properties

Class 11 Mathematics Quick Reference Sheet

1. The Fundamental Definition

Exponential Form	Logarithmic Form	Conditions
$a^x = b$	$\log_a b = x$	$a > 0, a \neq 1, b > 0$

2. Standard Laws of Logarithms

Rule Name	Mathematical Formulation
Product Rule	$\log_a (m \cdot n) = \log_a m + \log_a n$
Quotient Rule	$\log_a (m / n) = \log_a m - \log_a n$
Power Rule	$\log_a (m^k) = k \cdot \log_a m$

3. Change of Base Formulas

Property	Mathematical Formulation
Base Change Rule	$\log_a b = (\log_c b) / (\log_c a)$
Reciprocal Rule	$\log_a b = 1 / (\log_b a)$
Base Power Rule	$\log_{a^q} b = (1 / q) \cdot \log_a b$
Combined Power Rule	$\log_{a^q} (b^p) = (p / q) \cdot \log_a b$

4. Important Identities

Identity Property	Mathematical Formulation
Logarithm of Unit	$\log_a 1 = 0$
Logarithm of Base	$\log_a a = 1$
Self-Canceling Identity	$a^{\log_a x} = x$
Base-Switch Identity	$a^{\log_c b} = b^{\log_c a}$

5. Systems of Logarithms

Logarithm Type	Base	Standard Notation
Common Logarithm	Base 10	$\log_{10} x$ or $\log x$
Natural Logarithm	Base e ($e \approx 2.718$)	$\log_e x$ or $\ln x$